AI-POWERED REAL-TIME EXAM INVIGILATION SUPPORT SYSTEM USING YOLO AND BYTE TRACK

CAPSTONE PROJECT REPORT

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DECLARATION

We affirm that the project work titled **"AI-POWERED REAL-TIME EXAM INVIGILATION SUPPORT SYTEM USING YOLO AND BYTE TRACK"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

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PROJECT SUMMARY

Project Title	AI-POWERED REAL-TIME EXAM INVIGILATION SUPPORT SYSTEM USING YOLO AND BYTE TRACK
Project Team Members (Name with Register No)	Bhavana.Jasti(99220040547), Rushitha.Kuchi (99220040589),Venkataramanaiah.Gali(99220040509),Chakri Sriram.Dheekonda(99220040493).
Guide Name/Designation	Dr.M.K.Nagarajan , Associate Professor , Department of Computer Science and Engineering
Program Concentration Area	Artificial Intelligence with Deep Learning and Computer Vision
Technical Requirements	Python, Machine Learning & Deep Learning

Engineering standards and realistic constraints in these areas

Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	The system is cost-effective as it uses existing camera infrastructure and open-source software tools. This reduces deployment and maintenance expenses for educational institutions.	✓
Social	The system supports fair examination practices by assisting invigilators in real-time monitoring. It improves transparency without disrupting normal exam activities.	✓
Ethical	The system is designed with privacy and responsible AI usage considerations. Video data is processed strictly for examination supervision purposes.	✓
Sustainability	SDG 4:Quality Education	✓

REALISTIC CONSTRAINTS:

Economic

The system is economically efficient as it leverages existing camera infrastructure and open-source software, reducing the need for significant additional investment. It minimizes reliance on expensive hardware and licensed tools, thereby lowering both deployment and maintenance costs. By automating monitoring processes, it reduces manual supervision and recurring labor expenses while improving overall productivity. Additionally, the system optimizes resource utilization and minimizes operational inefficiencies, leading to long-term financial savings. Its scalability allows institutions to expand usage across multiple examination halls without a proportional increase in cost, making it a sustainable and cost-effective solution for educational organizations.

Social

From a social perspective, the system promotes fairness and equality in examinations by enabling real-time monitoring and ensuring all students are observed uniformly, which reduces malpractice and bias. It helps maintain a smooth and disturbance-free examination process while building trust and transparency among students and institutions. By fostering integrity, accountability, and equal opportunity, the system creates a level playing field for all learners and enhances institutional credibility. Additionally, it reduces the burden on invigilators through intelligent monitoring support, allowing them to focus on maintaining a calm and effective examination environment.

Ethical

The system is designed with strong ethical considerations, focusing on privacy and the responsible use of artificial intelligence. Video data is used strictly for examination supervision, with safeguards in place to protect student identity and prevent misuse. It emphasizes transparency, accountability, and fairness by operating within clearly defined boundaries and limiting surveillance to exam-related activities. Additionally, the system can incorporate consent mechanisms and data governance policies to ensure awareness among stakeholders, while adopting unbiased and accurate AI practices to uphold ethical standards and academic integrity.

Sustainability

The system supports sustainability by reducing reliance on physical resources such as additional invigilators and paper-based monitoring methods through a digital and automated approach. This minimizes resource consumption and lowers the environmental footprint by decreasing dependency on traditional, resource-intensive practices. Its technology-driven design allows easy updates and scalability without requiring extra physical resources, promoting efficient operations. Overall, the system encourages the adoption of smart, eco-friendly solutions in education, ensuring long-term sustainability aligned with modern environmental and institutional goals.

ABSTRACT

Cheating during examinations poses a serious threat to academic integrity and fairness in educational institutions. This project presents an AI-powered real-time exam invigilation support system designed to enhance monitoring efficiency in examination environments using advanced computer vision techniques. The system employs the YOLOv8 (You Only Look Once) object detection algorithm to identify students, invigilators, and prohibited items such as mobile phones and unauthorized notes within the examination hall. To ensure continuous monitoring, Byte Track is integrated for multi-object tracking, enabling persistent identification of individuals across consecutive video frames.

By combining rapid object detection with robust tracking, the system analyzes behavioral patterns such as frequent head turning, suspicious hand movements, object exchange, and abnormal proximity between candidates. Live or recorded video footage serves as input, and the system processes frames in real time to detect potential malpractice activities.

A web-based IP camera interface streams live video to a monitoring dashboard, where suspicious students are highlighted with red bounding boxes and normal behavioral is indicated with green bounding boxes. The proposed solution operates in real time with minimal human intervention, improving scalability, reducing monitoring errors, and supporting fair, reliable, and automated exam supervision.

Keywords— AI-Powered Invigilation, YOLO, Byte Track, Object Detection, Multi-Object Tracking, Suspicious Behavior Monitoring, Quality Education (SDG-4), Industry Innovation Infrastructure (SDG-9), Peace justice and Strong Institution (SDG-16).

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LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1.	211CSE1402	Python Programming
2.	212CSE2304	Machine Learning
3.	213CSE3301	Deep Learning

CHAPTER – I

INTRODUCTION

1.1 Background and Motivation

With the increasing reliance on technology in education, ensuring fair and transparent examination systems has become a major concern for institutions. Traditional examination monitoring methods often depend heavily on human invigilators, which can lead to inconsistencies, human error, and difficulty in managing large examination halls. The growing need for academic integrity and efficient supervision has motivated the development of intelligent systems that can assist in monitoring and detecting suspicious activities. Advances in artificial intelligence, computer vision, and automation have created opportunities to design smart solutions that enhance the effectiveness of examination processes while reducing manual effort. In recent years, the rapid digital transformation in the education sector has highlighted the importance of adopting smarter and more reliable examination management systems.

Conventional supervision methods often struggle to handle large groups of students efficiently, making it challenging to maintain consistent vigilance throughout the examination period. Limitations such as fatigue, limited attention span, and subjective judgment can affect the effectiveness of human monitoring. As educational institutions continue to grow and the demand for credibility in assessments increases, there is a strong need for innovative approaches that can support and enhance existing processes. Emerging technologies, particularly in artificial intelligence and automated visual analysis, provide the capability to continuously observe, analyzes, and interpret student behavioral with greater accuracy. These advancements enable the creation of intelligent monitoring solutions that not only strengthen exam security but also improve overall efficiency and reliability, ensuring a more standardized and trustworthy evaluation system.

1.2 Problem Statement

Despite the presence of invigilators, many examination environments still face challenges such as cheating, lack of proper supervision, and human limitations in continuously monitoring multiple students. Manual monitoring can be inefficient, prone to bias, and difficult to scale in large institutions. There is a need for a reliable and automated system that can assist in real-time monitoring, accurately detect unusual behavioral, and ensure fairness without disrupting the examination process. In many educational settings, maintaining strict discipline and fairness during examinations remains a significant challenge due to the limitations of traditional supervision methods. Human invigilators often find it difficult to maintain constant attention across large groups of students, which can create opportunities for unnoticed misconduct or inconsistencies in monitoring. Additionally, variations in individual judgment and the inability to track multiple activities simultaneously can affect the overall reliability of the examination process. As institutions expand and student populations grow, these challenges become even more pronounced, highlighting the need for a more dependable and scalable approach. Therefore, there is a growing demand for an intelligent system that can provide continuous observation, identify irregular patterns of behavioral, and support invigilators with

timely insights, ultimately ensuring a more secure, unbiased, and efficient examination environment.

1.3 Objectives of the Project

The primary objective of this project is to develop an intelligent examination monitoring system that enhances fairness, transparency, and efficiency. The system aims to assist invigilators by providing real-time monitoring and alerts for suspicious activities. It also focuses on reducing human workload, minimizing errors, and improving the overall integrity of the examination process. Additionally, the project seeks to create a cost-effective and scalable solution that can be easily adopted by educational institutions. The main aim of this project is to design and implement a smart and reliable system that improves the overall quality and credibility of examination monitoring. It focuses on integrating advanced technologies to support continuous observation and intelligent analysis of student behavioral during exams. By providing timely notifications and insights, the system helps authorities take quick action against any irregularities, thereby strengthening the examination process. Another key goal is to simplify supervision by reducing the burden on human invigilators and enhancing accuracy in monitoring. The project also emphasizes developing a flexible and efficient solution that can be easily adapted to different institutional requirements while maintaining affordability and ease of use.

1.4 Scope of the Project

The scope of the project includes the design and implementation of a system that uses camera-based monitoring and AI techniques to observe student behavioral during examinations. It focuses on detecting anomalies such as unusual movements or suspicious actions within the examination hall. The system is intended for use in controlled environments like classrooms or examination halls and does not extend to remote or online examination scenarios. It emphasizes real-time monitoring and reporting while ensuring privacy and ethical considerations are maintained. The project is centered on developing a technology-driven solution that can monitor and analyze student activities within a structured examination environment using visual data and intelligent algorithms. It primarily focuses on identifying patterns of behavioral that may indicate irregular or unauthorized actions, enabling timely intervention when necessary. The implementation is designed for physical examination settings where controlled conditions allow for accurate data capture and analysis.

The system is intended to function as a supportive tool for invigilators, enhancing their ability to oversee large groups effectively without interfering with the normal conduct of exams. Additionally, the project considers important aspects such as data handling, user accessibility, and system reliability, ensuring that the solution operates efficiently while adhering to necessary ethical standards and institutional requirements. It operates as an assistive tool that complements the role of invigilators by enhancing their ability to manage large groups of students efficiently without causing disruption. Furthermore, the project emphasizes robustness, usability, and secure data management, ensuring that the system delivers reliable performance while aligning with institutional policies, ethical guidelines, and privacy considerations.

1.5 Methodology Overview

The project adopts a well-structured and iterative methodology to ensure the development of an effective and reliable examination monitoring system. It begins with the collection of relevant visual data from examination settings, followed by pre-processing steps such as cleaning, normalization, and feature extraction to prepare the data for analysis. Advanced techniques from machine learning and computer vision are then utilized to build models capable of recognizing patterns and identifying abnormal or suspicious behavioral. These models are integrated into a functional system that includes a user-friendly interface, enabling invigilators to observe activities and receive instant notifications when irregularities are detected. The methodology also incorporates continuous testing, validation, and refinement to improve system performance, ensuring high accuracy, efficiency, and robustness in real-world scenarios.

1.6 Organization of the Report

The report is organized into several chapters to clearly present the development and evaluation of the proposed system. **Chapter I** provides the introduction to the project, including the background and motivation behind the work, the problem statement, the objectives, scope, methodology overview, and the structure of the report. **Chapter II** presents the literature review, which discusses related studies, previously developed systems, and research papers relevant to the project, followed by a summary that highlights the research gaps addressed by the proposed system. **Chapter III** focuses on system analysis, where the project requirements are identified and categorized into functional and non-functional requirements. It also includes feasibility studies covering technical, operational, and economic aspects, along with risk analysis related to the system implementation. **Chapter IV** discusses the design alternatives and trade-off analysis, explaining the selection of the detection approach, video processing techniques, tracking mechanisms, decision-making strategies, and evidence management methods used in the system. **Chapter V** describes the system design and implementation in detail. It includes the overall architecture, video input and frame extraction process, object detection using YOLOv8, multi-object tracking using Byte Track, suspicious behaviour analysis module, ER diagram, database schema, development environment, YOLOv8 model training, and the real-time detection pipeline. **Chapter VI** explains the feedback and iterative refinement process, including feedback collection from users, evaluation of detection performance, model tuning, interface improvements, and continuous system validation. **Chapter VII** presents the testing and validation of the system, describing the testing environment, functional testing of different modules, performance testing of the detection model, validation of real-time monitoring and tracking, and testing of the evidence recording and alert system. **Chapter VIII** discusses the results obtained from the system, including output screenshots, evaluation metrics such as F1-confidence curve, precision-recall curve, confusion matrix, comparison with existing systems, challenges faced, and the solutions implemented to overcome them.

Finally, **Chapter IX** concludes the report by summarizing the overall outcomes of the project and discussing possible future enhancements. The report also includes references to the resources used, publication details, certification, plagiarism report, and a project QR code for quick access to project-related information.

CHAPTER-II

LITERATURE REVIEW

2.1 Overview of Related Work

The literature on intelligent examination monitoring systems highlights the growing use of advanced technologies such as artificial intelligence, machine learning, and computer vision to enhance supervision in academic environments. Many studies have explored automated surveillance systems capable of analyzing video streams to detect unusual patterns of behavioral, including head movements, object detection, and suspicious interactions among students. These systems aim to overcome the limitations of traditional invigilation by providing continuous and consistent monitoring. Researchers have also focused on improving accuracy, reducing false positives, and ensuring real-time performance, making such systems more reliable and practical for deployment in educational institutions.

Recent developments in educational technology have led to the emergence of smart monitoring solutions that aim to improve the efficiency and reliability of examination supervision. Various research efforts have focused on integrating intelligent systems that can continuously observe student activities and interpret behavioral patterns within exam settings. These approaches emphasize the use of automated analysis to assist human invigilators in maintaining discipline and identifying irregularities more effectively. Several studies also highlight the importance of real-time processing, enabling immediate detection and response to suspicious actions, which enhances the overall integrity of examinations. In addition, researchers have explored methods to make these systems more adaptable to different environments, ensuring consistent performance across varying classroom conditions. The ongoing advancements in this field demonstrate a clear shift towards technology-driven supervision, aiming to create a more secure, efficient, and standardized examination process while addressing the limitations of manual monitoring.

AI-Powered Classification for Cheating Detection in Offline Examinations

Rajasekar and Vijayarani introduced an automated invigilation framework that utilizes deep learning and computer vision to monitor examination environments. Their research focused on identifying suspicious student behaviors using object detection techniques. The system produced encouraging results under controlled laboratory conditions. However, its performance decreased in crowded examination halls, and the absence of persistent multi-object tracking limited the system's ability to continuously monitor student activities across video frames [1].

AI-Based Exam Proctoring System Using YOLO

Nagajothi et al. proposed a prototype AI-based proctoring system that employed the YOLO object detection algorithm to identify students and restricted items within examination halls. The framework also incorporated facial recognition to verify student attendance. Although the system demonstrated effective detection capabilities, it relied on a single-camera setup and lacked advanced behavioral analysis over time. Continuous identity tracking was not extensively implemented, which reduced its reliability in dynamic examination environments [2].

Proctoring Systems Using Artificial Intelligence and Machine Learning: A Review
Ghita et al. conducted a comprehensive survey of AI-based proctoring systems developed between 2010 and 2021. The review classified various approaches such as object detection, gaze tracking, and behavioral analysis methods. While the survey provided valuable insights into system design and evaluation strategies, many of the systems reviewed lacked scalability for real-time monitoring and did not incorporate robust tracking mechanisms required for large examination halls [3].

AI-PoweredOnlineExamProctoringSystemUsingDeepLearning
Kumar et al. proposed an online examination proctoring system that integrates deep learning with computer vision techniques. Their approach utilized multimodal data sources, including video and audio streams, to detect suspicious student behavior. Although the system achieved improved detection accuracy compared to conventional methods, it produced false positives under inconsistent lighting conditions and required significant computational resources, limiting its practical deployment in offline classroom environments [4].

SmartProctoringSystemUsingYOLOv8
Kaur et al. developed a smart proctoring system using an enhanced YOLOv8 model to detect abnormal behaviors during examinations. The model achieved higher object detection accuracy; however, it required extensive fine-tuning for different classroom layouts and environments. The framework primarily emphasized detection performance and did not focus on persistent identity tracking, which is essential for long-term monitoring in real-time examination settings [5].

Real-TimeClassroomMonitoringUsingYOLOandLSTM
Reddy et al. implemented a real-time monitoring system that combined YOLO for object detection with Long Short-Term Memory (LSTM) networks for analyzing sequential behavioral patterns. This integration aimed to detect suspicious actions over time rather than in isolated frames. Despite improvements in behavioral recognition, the system experienced generalization challenges due to limited dataset diversity and faced difficulties in handling occlusions in crowded classrooms [6].

Suardana et al. applied the YOLOv5 model to detect cheating-related movements in offline examination environments and reported promising accuracy results. However, the system required extensive dataset preparation and a large number of annotated samples. Additionally, persistent identity tracking was not fully addressed, which reduced system robustness under dynamic conditions [7].

Improved YOLOv8 with Attention Mechanism for Abnormal Behavior Detection
Wang and Chen proposed an improved YOLOv8 architecture integrated with attention mechanisms to enhance abnormal behavior detection in examination halls. The model demonstrated improved detection accuracy but relied heavily on high-quality camera input and controlled environmental conditions. Performance degradation was observed under low-light environments and partial occlusion scenarios [8].

MultimodalCheatingDetectionFramework
Ahmed and Rao developed a multimodal proctoring system that combined YOLO-based face detection, gaze tracking, and audio analysis to detect various forms of cheating. The system was capable of identifying both visual and speech-based malpractice. However, the framework required substantial computational power and complex infrastructure, making it difficult to deploy in real-time offline examination settings [9].

PoseEstimation-BasedExaminationMonitoringSystem
Banerjee and Mehta incorporated pose estimation techniques alongside YOLO-based detection to recognize signaling gestures among students during examinations. While the system improved gesture-based suspicious activity detection, its performance declined when students were seated close together or when occlusions occurred frequently [10].

Holi et al. proposed an intelligent offline examination monitoring system that utilized convolutional neural networks to classify student behavior into multiple categories. The system achieved high accuracy under controlled testing environments. However, the dataset used for training was relatively small, and the model's effectiveness in real-time crowded examination scenarios was not extensively validated [11].

Real-Time Detection of Examination Malpractices Using CNN and Video Surveillance
Adeyemi et al. conducted a systematic review and meta-analysis of CNN-based examination monitoring systems. Their findings highlighted the effectiveness of deep learning models in detecting malpractice activities through video surveillance. However, many of the reviewed approaches lacked persistent multi-object tracking and struggled with environmental challenges such as occlusion and lighting variations, limiting their scalability for large examination halls [12].

Implementation of Intelligent Exam Supervision System Using Deep Learning
Mahmood et al. proposed an intelligent examination supervision system based on deep learning techniques. The system used object detection algorithms to identify suspicious student movements and unauthorized objects during examinations. Although promising results were reported in controlled conditions, the system required high computational resources and did not implement stable identity tracking across video frames [13].

MultipleInstanceLearningforCheatingDetectionandLocalization
Liu et al. introduced a cheating detection framework based on Multiple Instance Learning (MIL) for online examination environments. The model focused on localizing suspicious activities within video segments rather than detecting them frame by frame. While the approach improved activity localization, it did not include real-time tracking mechanisms, making it less suitable for continuous offline examination monitoring [14].

Zhang et al. explored the use of reinforcement learning for stock prediction and high-frequency trading systems operating under T+1 trading rules. The study proposed a dynamic parameter optimization approach to improve stock price trend forecasting and trading decisions. However, the training process required large datasets and significant computational power, which limits real-time deployment in practical scenarios [14].

In-Class Exam Auto Proctoring Using Deep Learning on Student Behaviors Radwan et al. proposed an automated proctoring system that analyzes student behavioral patterns during in-class examinations using deep learning models. The system detected abnormal head movements and interaction behaviors among students. However, the detection accuracy decreased when students were partially occluded or when camera placement varied significantly [15].

AI-Powered Classification for Cheating Detection in Offline Examinations Jahangir et al. introduced a deep learning-based classification model designed to detect cheating activities in offline examination environments. The system relied on image-based classification trained on a custom dataset. Although the model achieved satisfactory classification accuracy, it lacked integrated multi-object tracking capabilities and did not address continuous identity monitoring in real-time scenarios [16].

2.2 Review of Similar Projects or Research Papers

Several projects and research papers have proposed solutions for automated exam monitoring using various approaches. Some systems utilize facial recognition and gaze tracking to identify inattentive or suspicious behavioral, while others employ deep learning models such as Convolutional Neural Networks (CNNs) for activity recognition. Certain studies have integrated multiple techniques, combining motion detection, object recognition, and behavioral analysis to improve system performance. Additionally, a few solutions are designed for online examinations, incorporating webcam monitoring and browser activity tracking. While these approaches demonstrate promising results, they often face challenges such as high computational requirements, limited accuracy in complex environments, and concerns related to privacy and data security.

A wide range of existing systems and research efforts have explored automated solutions to enhance examination monitoring by applying different technological strategies. Many of these works focus on analyzing visual inputs to detect student activities and identify irregular patterns that may indicate misconduct. Some approaches rely on tracking facial features and eye movements to assess attention levels, while others use advanced learning algorithms to classify actions based on observed behavioral. In several cases, hybrid models have been developed that combine multiple techniques to achieve better accuracy and robustness in detection. There are also solutions tailored for digital or remote examinations, where monitoring is carried out through webcams and system activity analysis. Although these methods show significant potential in improving exam supervision, they often encounter limitations such as high processing demands, reduced effectiveness in crowded or dynamic environments, and challenges in maintaining user privacy and data protection standards.

2.3 Summary and Gap Identification

From the analysis of existing literature, it is evident that although significant progress has been made in automated examination monitoring, several gaps still remain. Many systems are either too complex or resource-intensive, making them difficult to implement in real-world educational settings. Some solutions lack scalability or fail to perform effectively in crowded examination halls. Additionally, there are concerns regarding ethical usage, data privacy, and system reliability. These limitations highlight the need for a more balanced solution that is accurate, cost-effective, scalable, and ethically designed. The proposed project aims to address these gaps by developing a system that combines efficient monitoring techniques with practical implementation, ensuring both effectiveness and usability in real examination environments.

Many current solutions focus heavily on achieving high accuracy but often overlook factors such as ease of deployment, adaptability, and user convenience in real institutional environments. In several cases, systems struggle to maintain consistent performance when applied to large-scale settings or diverse classroom conditions. Additionally, concerns related to transparency, data handling, and trustworthiness continue to influence the acceptance of such technologies. These observations indicate the necessity for a more well-rounded approach that balances technical performance with simplicity, efficiency, and responsible usage. Therefore, there is a clear need to develop a solution that not only delivers reliable monitoring capabilities but also ensures practical usability, scalability, and adherence to ethical standards within real-world examination scenarios.

Another important limitation observed in many existing systems is the lack of effective multi-object tracking and continuous behavior analysis. Most solutions focus primarily on detecting objects or specific actions within individual frames, but they do not maintain consistent identity tracking of students across multiple frames. This limitation reduces the system's ability to accurately monitor long-term behavior patterns and may lead to incorrect or incomplete detection of suspicious activities. Furthermore, several proposed frameworks require high computational resources or complex infrastructure, which makes their implementation difficult in institutions with limited technical facilities.

In addition, scalability and adaptability remain key challenges. Many systems are designed and tested under controlled environments and may not perform consistently in crowded examination halls where multiple students, varying seating arrangements, and different camera angles are involved. These challenges highlight the need for a more practical and efficient monitoring solution that can operate reliably in real-time conditions.

Therefore, the proposed project aims to address these research gaps by developing an AI-powered exam invigilation support system that integrates efficient object detection and robust tracking mechanisms. By utilizing modern deep learning models and lightweight tracking algorithms, the system seeks to provide accurate monitoring, improved identity tracking, and better adaptability to real examination environments while maintaining computational efficiency.

CHAPTER-III

SYSTEM ANALYSIS

3.1Requirements Gathering

The requirements gathering process involves a detailed study of examination environments to understand the practical challenges faced during monitoring and supervision. It focuses on identifying the expectations of key stakeholders such as invigilators and institutional authorities while analyzing existing examination procedures. By evaluating the limitations of manual invigilation, such as difficulty in monitoring large groups and identifying suspicious behavioral, the system requirements are carefully defined. This process also includes observing real-time exam scenarios and reviewing current practices to ensure that the proposed solution addresses real-world needs effectively. The outcome is a set of well-defined, clear, and relevant requirements that guide the development of a user-friendly, efficient, and adaptable system.

3.2Functional Requirements

The functional requirements describe the essential operations that the system must perform to ensure effective examination monitoring. The system is designed to capture live video input and process it using intelligent algorithms to analyze student behavioral in real time. It should be capable of detecting and classifying activities, assigning unique identities to individuals through tracking, and maintaining records of their actions. Additionally, the system generates alerts when suspicious behavioral is detected, allowing invigilators to take immediate action. It also supports storing video evidence and providing a dashboard interface where users can monitor live activity and review status updates. These functionalities collectively enhance supervision and support accurate and timely decision-making.

3.3 Non-Functional Requirements

The non-functional requirements define the quality and performance standards that the system must meet to operate effectively. The system should deliver high accuracy in detection while maintaining minimal latency to support real-time monitoring. It must be reliable enough to function continuously during examination sessions without interruptions or failures. Scalability is another important aspect, ensuring that the system can handle varying numbers of students and different examination hall sizes. Security and privacy are critical, requiring that all collected data is protected and used only for authorized purposes. In addition, the system should be easy to use, with an intuitive interface that allows invigilators to operate it efficiently without requiring advanced technical skills.

3.4 Feasibility Study

The feasibility study evaluates the practicality of implementing the proposed system by analyzing various constraints and available resources. It helps determine whether the system can be successfully developed and deployed within the given conditions. This evaluation considers technical capabilities, operational suitability, and economic factors to ensure that the project is viable. By examining these aspects, the feasibility study provides confidence that the system can be implemented effectively and will deliver the expected outcomes without significant limitations.

3.4.1TechnicalFeasibility

The system is technically feasible as it utilizes widely available and proven technologies such as computer vision, machine learning models, and web-based frameworks. The use of existing camera infrastructure and open-source tools simplifies the development process and reduces technical complexity. Modern computing systems are capable of handling real-time video processing and analysis, making it possible to implement the system efficiently. These factors confirm that the project can be developed using current technological resources without requiring specialized or unavailable components.

3.4.2 OperationalFeasibility

From an operational perspective, the system is designed to integrate seamlessly into existing examination processes. It acts as a supportive tool for invigilators, enhancing their monitoring capabilities rather than replacing them. The system provides real-time alerts and an easy-to-use interface, making it convenient for users to adopt with minimal training. Its implementation does not disrupt the normal flow of examinations, ensuring smooth operation. This makes the system practical and acceptable for use in real educational environments.

3.4.3 Economic Feasibility

The system is economically feasible as it minimizes costs by leveraging existing infrastructure such as cameras and utilizing open-source software tools. This reduces the need for additional investments in hardware and licensing. By automating monitoring processes, the system also decreases reliance on manual supervision, leading to long-term cost savings. Its scalability allows institutions to expand usage without significant increases in expenditure, making it a cost-effective solution suitable for institutions with varying financial capabilities.

3.5 Risk Analysis

Risk analysis involves identifying potential challenges that may arise during the development and deployment of the system and preparing strategies to mitigate them. Possible risks include inaccuracies in detection due to environmental factors, system performance limitations, and issues related to data privacy and security. There may also be challenges in user acceptance and system integration within existing workflows. To address these risks, the system can be improved through continuous testing, model optimization, secure data handling practices, and proper user training. By proactively managing these risks, the system can achieve reliable performance and maintain effectiveness in real-world examination scenarios.

CHAPTER-IV

DESIGN ALTERNATIVES AND TRADE-OFF ANALYSIS

4.1 Detection Approach Selection

The selection of an appropriate detection approach is a critical aspect of the system design, where both traditional methods and deep learning-based techniques are carefully evaluated for identifying student behavioral. Traditional approaches, such as rule-based or basic feature extraction methods, are relatively simple to implement and require lower computational resources, making them suitable for basic applications. However, they often struggle to perform effectively in complex and dynamic environments like examination halls, where variations in lighting, movement, and student interactions can affect accuracy. In contrast, deep learning models offer advanced capabilities in recognizing patterns and handling multiple classes of behavioral, such as distinguishing between normal activity, suspicious actions, and the presence of invigilators. These models are more adaptable to real-world conditions and provide higher detection accuracy. Although they require greater computational power and a well-prepared training dataset, the improved performance and reliability they offer make them a more suitable choice. Therefore, the system adopts a deep learning-based approach to ensure accurate, consistent, and real-time detection in examination environments.

4.2 Video Processing Method

An important design decision in the development of the system is the choice between static image analysis and real-time video processing for monitoring student behavioral. Static image analysis offers simplicity in implementation and requires relatively lower computational resources, as it processes individual frames independently. However, it is limited in its ability to capture continuous actions, temporal changes, and behavioral patterns that occur over time, which are crucial for accurately identifying suspicious activities in examination settings. In contrast, real-time video processing enables continuous observation of students, allowing the system to analyze sequences of actions and detect irregular behavioral as it happens. This approach provides a more comprehensive understanding of student activities and supports immediate response to potential misconduct. Although real-time processing introduces higher computational demands and increases system complexity, its ability to deliver accurate, dynamic, and timely monitoring makes it a more effective and appropriate choice for examination environments.

A key design choice involves selecting between static image analysis and real-time video processing. Static image processing is easier to implement but cannot capture continuous actions or behavioral patterns. Real-time video processing allows the system to monitor students continuously and detect suspicious activities as they occur. While this increases system complexity and processing requirements, it significantly enhances monitoring effectiveness, making it more suitable for examination environments.

4.3 Tracking Mechanism

The tracking mechanism is an essential design consideration in the system, where the choice is made between simple detection and object tracking across multiple frames. Detection-only

approaches analyze each frame independently and are unable to maintain consistency in identifying individuals over time, which can lead to confusion and misinterpretation of student behavioral. This limitation makes it difficult to accurately monitor activities in a continuous and reliable manner. To overcome this, the system incorporates object tracking, which assigns a unique identifier to each student and follows their movements across frames. This enables continuous observation and allows the system to build a behavioral history for each individual. As a result, it becomes possible to analyze patterns over time and implement advanced features such as suspicion scoring for more accurate decision-making. Although the inclusion of tracking increases computational complexity and processing requirements, the significant improvement in accuracy, consistency, and overall monitoring effectiveness makes it a valuable and necessary component of the system.

4.4 Decision-Making Strategy

The decision-making strategy of the system involves a careful comparison between instant detection and a cumulative scoring approach to identify suspicious behavior effectively. Instant decision-making methods can quickly flag activities as abnormal based on a single observation, making them fast and responsive. However, such approaches are often prone to inaccuracies because temporary misclassifications or brief, non-suspicious actions may be incorrectly interpreted as cheating. To overcome this limitation, the system adopts a cumulative suspicion scoring mechanism, where observations are collected and analyzed over a period of time rather than relying on a single frame. By evaluating repeated behavior patterns, the system ensures that decisions are more reliable and less influenced by momentary errors. This approach significantly reduces false positives and improves the overall accuracy of detection. Although it introduces a slight delay in generating final alerts, the increased reliability and fairness in decision-making make it a more effective strategy for real-world examination monitoring systems.

4.5. Evidence Management

The evidence management strategy in the system involves evaluating the trade-off between continuous video recording and selective evidence storage to ensure efficient use of resources. Continuous recording captures the entire examination session, providing complete coverage of all activities, but it requires a large amount of storage space and often includes unnecessary data that may not be relevant for monitoring purposes. To address this limitation, the system adopts a selective evidence storage approach using a frame buffer mechanism. This technique temporarily stores recent video frames and saves them as evidence only when suspicious activity exceeds a predefined threshold, ensuring that only important events are recorded. This approach significantly reduces storage requirements while still preserving critical information needed for review and analysis. However, it may result in some minor or brief incidents not being captured. Despite this trade-off, the method provides an efficient and practical solution for managing evidence in real-time examination monitoring systems.

CHAPTER-V

SYSTEM DESIGN AND IMPLEMENTATION

5.1 Overall System Architecture

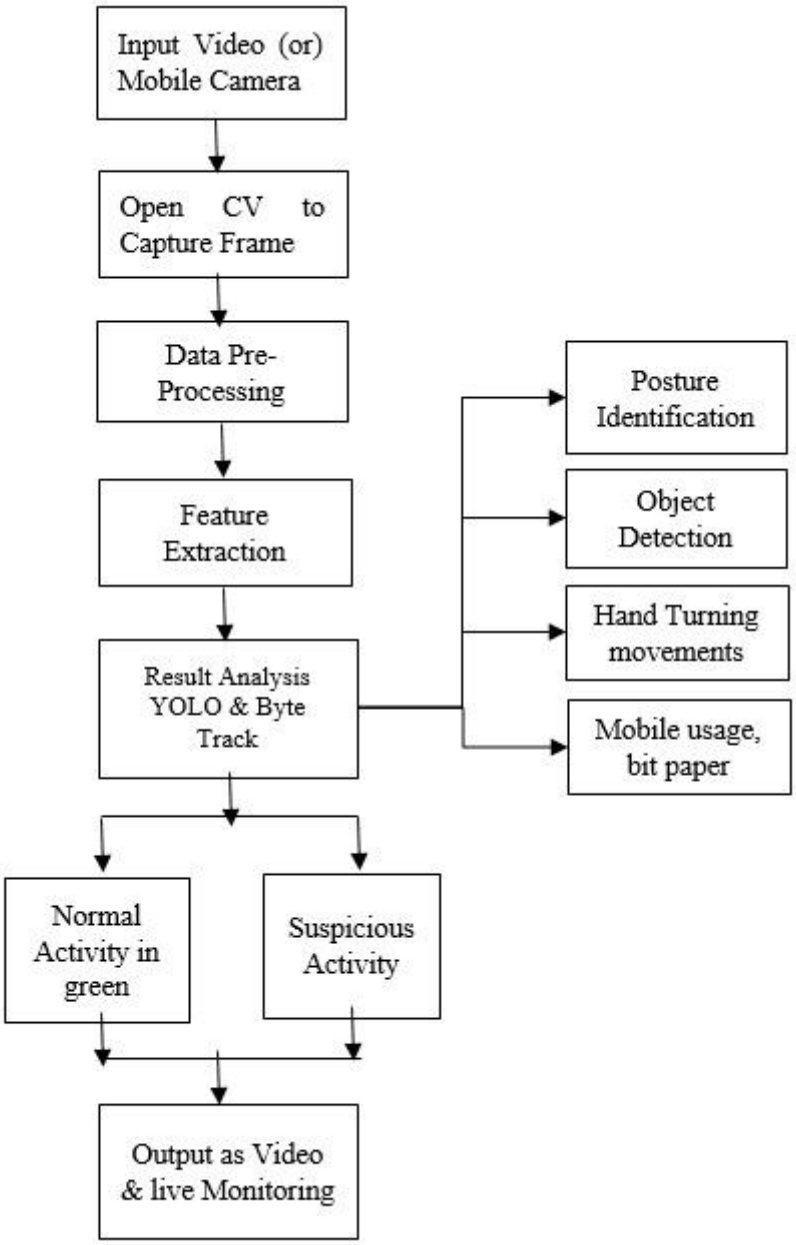

Fig-1: System Architecture

The overall system architecture of the proposed examination monitoring system is designed as an integrated pipeline that combines video input, intelligent processing, and user interface components to ensure efficient real-time monitoring. The system begins with capturing live video data from a camera source, which is then processed frame by frame using advanced computer vision techniques. The architecture includes modules for object detection, tracking, behavior analysis, and alert generation, all working together in a coordinated manner. The processed output is displayed through a web-based dashboard, allowing invigilators to monitor activities and receive updates in real time. This modular and scalable architecture ensures flexibility, easy maintenance, and the ability to extend the system with additional features in the future.

5.2 Video Input and Frame Extraction Module

The video input and frame extraction module is responsible for capturing live video streams from the camera and converting them into individual frames for further processing. This module ensures that the input video is continuously read and resized to a suitable resolution for efficient computation. Each frame is extracted in real time and passed to subsequent modules for analysis. The design of this module focuses on maintaining a steady frame rate and minimizing delays, ensuring smooth and uninterrupted data flow. It plays a crucial role in enabling real-time performance and forms the foundation for all further processing steps in the system. The video input and frame extraction module plays a fundamental role in the overall functioning of the system by acting as the entry point for all visual data processing.

It is responsible for continuously capturing live video streams from the camera source and efficiently converting the incoming stream into a sequence of individual frames that can be analyzed by subsequent modules. This module ensures that the captured video is processed in a consistent and structured manner by resizing frames to an optimal resolution, which helps in balancing computational efficiency and detection accuracy. It is designed to handle real-time data flow without interruptions, maintaining a stable frame rate to ensure that no critical information is lost during processing. Additionally, the module minimizes latency by quickly extracting and forwarding frames to the object detection and tracking components, enabling smooth and seamless operation of the entire system. By ensuring a continuous and reliable supply of processed frames, this module forms the backbone of the real-time detection pipeline and significantly contributes to the overall performance, responsiveness, and effectiveness of the examination monitoring system

5.3 Object Detection Module (YOLOv8 Integration)

The object detection module utilizes the YOLOv8 model to identify and classify objects present in each frame. This module is trained to detect specific classes such as normal students, suspicious behavior, and invigilators. By applying deep learning techniques, the system can accurately recognize patterns and differentiate between various activities in the examination environment. The integration of YOLOv8 enables fast and efficient detection with high accuracy, making it suitable for real-time applications. This module serves as the core

component of the system, providing the necessary information for further analysis and decision-making. This module is specifically designed to detect and classify relevant entities in the examination environment, such as normal student behavior, suspicious activities, and the presence of invigilators. By leveraging advanced deep learning techniques, the model is capable of learning complex visual patterns and distinguishing between different types of actions with high precision. The use of YOLOv8 ensures that detection is performed at high speed without compromising accuracy, which is essential for real-time monitoring scenarios. Each processed frame is analyzed to extract meaningful information, including object location, classification, and confidence levels, which are then forwarded to subsequent modules for further processing. The efficiency and reliability of this module significantly impact the overall performance of the system, as it forms the foundation for tracking, behavior analysis, and decision-making processes, ultimately enabling effective and intelligent examination monitoring.

Feature	Description
Class Label	Predicted category: Normal, Cheating, or Invigilator
Confidence Score	Probability value indicating detection reliability
Track ID	Unique identifier assigned to each individual by Byte Track
Frame Number	Sequential frame index used for temporal behavior analysis

Table 1: Input Features Used

5.4 Multi-Object Tracking Module (Byte Track Integration)

The multi-object tracking module is implemented using the Byte Track algorithm to maintain consistent identification of individuals across multiple frames. After detecting objects, this module assigns unique IDs to each student and tracks their movements throughout the video stream. This continuous tracking allows the system to monitor behavior over time rather than relying on single-frame observations. It ensures that each student's actions are accurately recorded and analyzed, enabling more reliable detection of suspicious patterns. Although tracking adds computational complexity, it significantly enhances the system’s ability to provide consistent and meaningful insights.After the object detection stage identifies students and other entities, this module assigns a unique ID to each individual and continuously tracks their position and movement throughout the video stream. This capability allows the system to maintain identity continuity, ensuring that each student is monitored accurately over time rather than being treated as a new object in every frame. By tracking movements and maintaining a history of actions, the system can analyze behavioral patterns more effectively and detect suspicious activities with greater reliability. This temporal consistency is crucial for implementing advanced features such as suspicion scoring and long-term behavior analysis. Although the integration of tracking increases computational complexity and processing requirements, it greatly enhances the overall accuracy, stability, and interpretability of the system, making it a vital component for real-time examination monitoring.

5.5 Suspicious Behaviour Analysis module

The suspicious behavior analysis module is responsible for interpreting detected and tracked data to identify potential misconduct. It evaluates patterns of behavior over time and calculates a suspicion score for each individual based on their actions. This module ensures that decisions are not made based on a single observation but rather on consistent patterns, reducing false positives. When the suspicion score exceeds a predefined threshold, the system triggers alerts and may initiate evidence recording. This intelligent analysis improves the reliability and fairness of the monitoring process. This module continuously evaluates the behavior of each tracked individual by analyzing patterns over a period of time rather than relying on isolated observations. It assigns and updates a suspicion score for every student based on their detected actions, allowing the system to differentiate between normal variations in behavior and consistent suspicious activity. By focusing on accumulated patterns, the module significantly reduces the chances of false positives that may arise from momentary or accidental movements. When the calculated suspicion score crosses a predefined threshold, the system takes appropriate actions such as generating alerts for invigilators and initiating the recording of evidence for further review. This approach ensures that decisions are data-driven, consistent, and fair, thereby enhancing the overall reliability and effectiveness of the examination monitoring system while supporting informed decision-making.

5.6 ER-Diagram

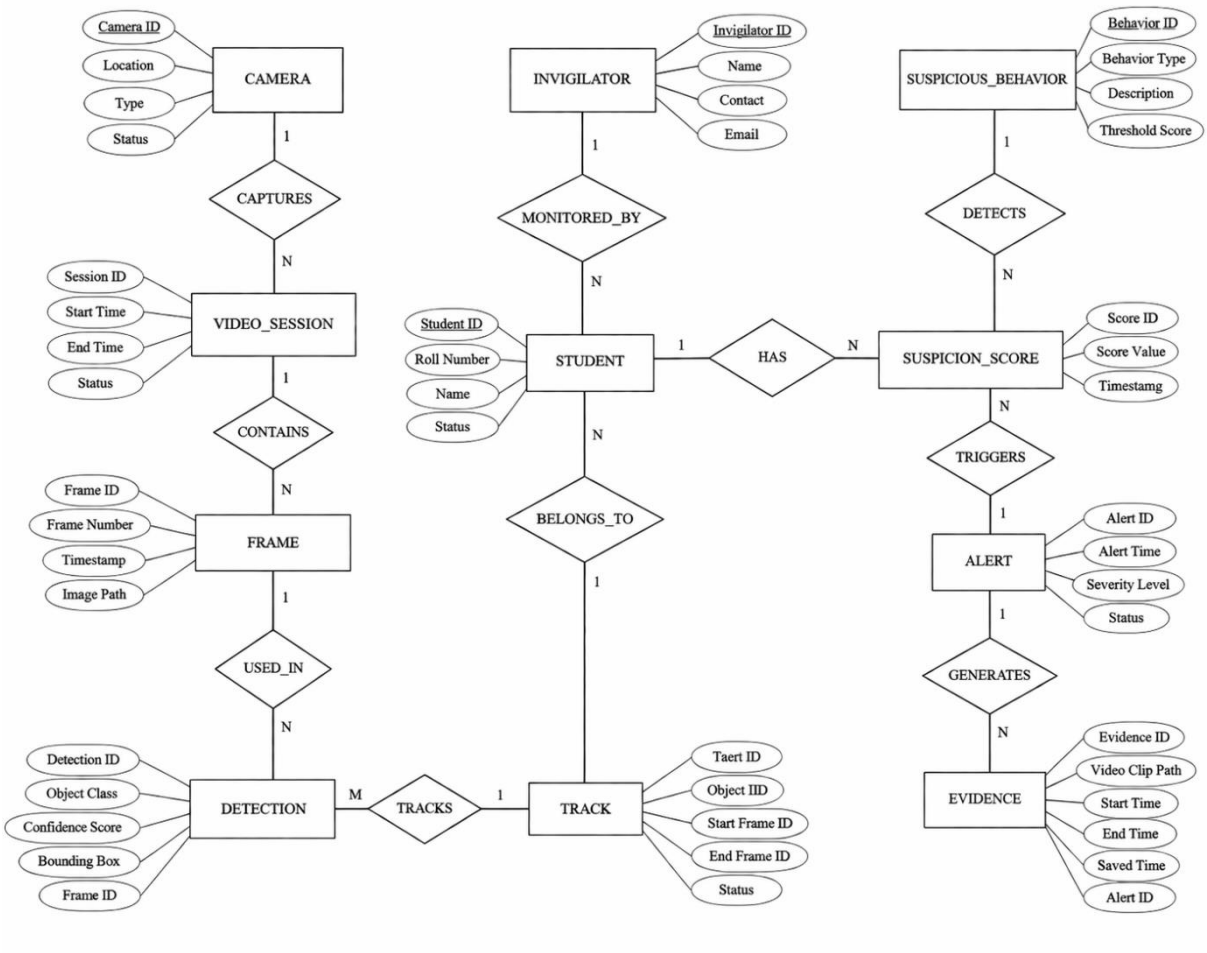

Fig-2: ER-Diagram of Real Time Exam Invigilation System

The ER diagram represents the structure and relationships of an intelligent examination monitoring system, showing how different components interact with each other. The main entities in the system include Camera, Video Session, Frame, Detection, Track, Student, Invigilator, Suspicious Behavior, Suspicion Score, Alert, and Evidence. The Camera entity captures video data and is connected to the Video Session entity through a captures relationship, indicating that one camera can capture multiple video sessions. Each Video Session contains multiple Frames, representing the continuous extraction of images from the video stream. These frames are then used in the Detection entity, where objects such as students or suspicious activities are identified. The Detection entity is linked to the Track entity through a tracks relationship, where multiple detections can belong to a single track. The Track entity maintains consistency by assigning a unique ID to each detected individual and tracking them across frames. Each track is associated with a Student, establishing a belongs to relationship, which ensures that the system monitors individual student behavior over time. The Student entity is also connected to the Invigilator through a monitored by relationship, indicating that one invigilator can oversee multiple students. To analyze behavior, the system includes a Suspicion Score entity, which is linked to the student through a has relationship. This score is calculated based on observed actions over time.

5.7 Database schema

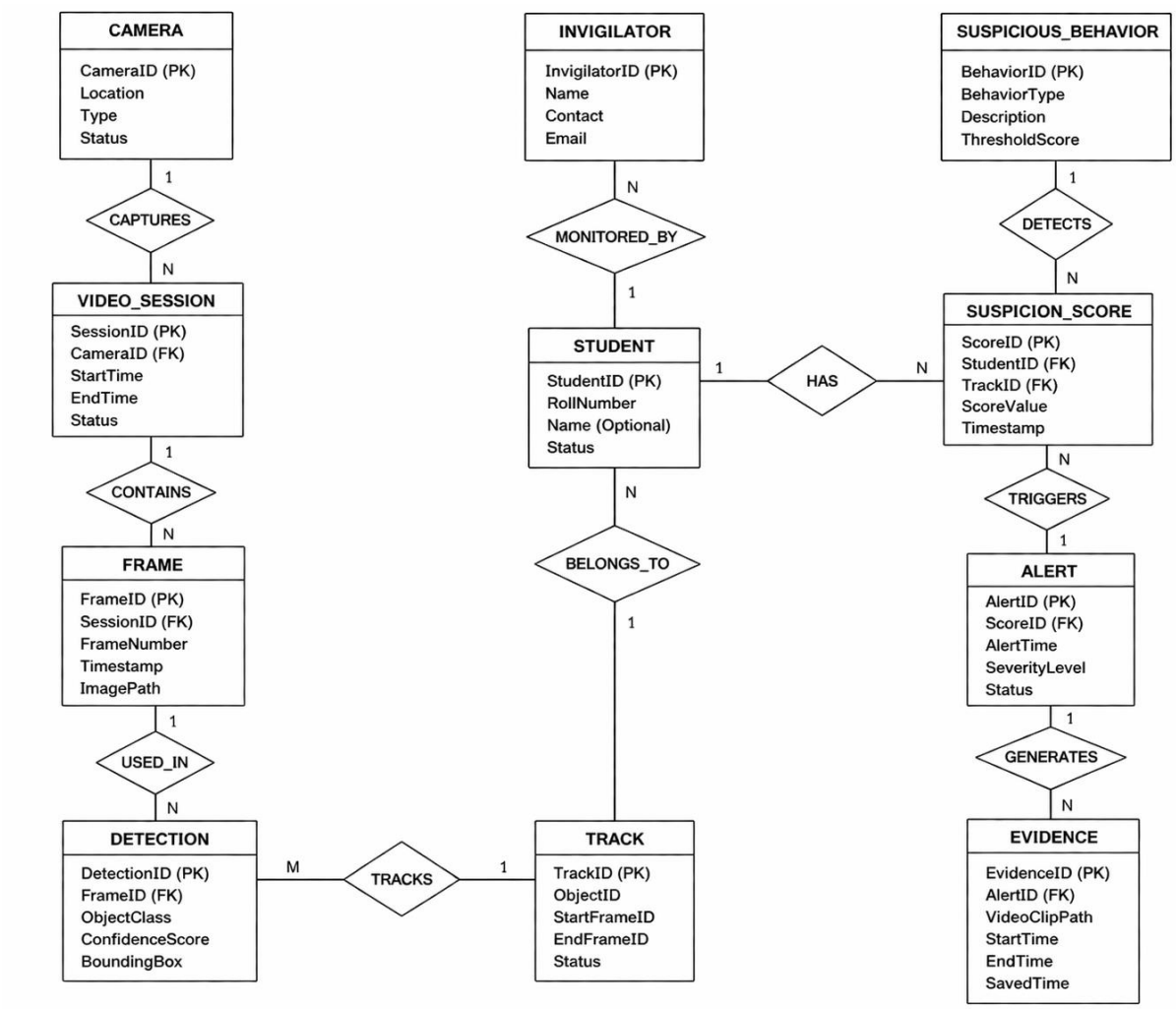

Fig-3: Database schema of Real Time Exam-Invigilation support system

The database schema of the intelligent examination monitoring system is designed to efficiently manage data related to video input, detection, tracking, and behavior analysis. It includes tables such as Camera, Video Session, and Frame to store video-related information, while Detection and Track tables handle object identification and continuous tracking of students. The Student and Invigilator tables maintain user details and supervision relationships. Behavioral analysis is managed using Suspicious Behavior and Suspicion Score tables, which evaluate and store student activity patterns over time. When suspicious activity is detected, the Alert table records warnings, and the Evidence table stores related video clips for verification. Overall, the schema ensures organized data storage, smooth data flow between modules, and supports accurate and real-time monitoring of examination environments.

5.8 Development Environment

The system was developed using Python as the primary programming language due to its extensive support for deep learning and computer vision libraries. The following tools and frameworks were used:

- **Ultralytics YOLOv8** for object detection model training and inference
- **Byte Track** for multi-object tracking
- **OpenCV** for video capture and frame processing
- **Flask** for real-time streaming and monitoring interface
- **NumPy** for numerical computations

The system was executed in an environment supporting GPU acceleration for faster inference, while also maintaining compatibility with CPU-based execution for practical deployment.

Component	Tool / Library
Programming Language	Python 3.9+
Object Detection Framework	Ultralytics YOLOv8
Multi-Object Tracking	Byte Track
Web Framework	Flask
Video Streaming	OpenCV + Flask Streaming
Visualization	OpenCV Drawing Functions

Table 2: Software Stack

5.9 YOLOv8 Model Training:

The YOLOv8 model was trained using a custom dataset containing annotated images categorized into three classes: Normal, Cheating, and Invigilator.

Training Steps:

1. Dataset preparation with bounding box annotations.
2. Configuration of the dataset in YAML format.
3. Selection of YOLOv8 model variant.
4. Training with specified epochs, batch size, and image size.
5. Monitoring loss curves and validation metrics.

The training process optimizes bounding box regression loss, classification loss, and objectiveness loss simultaneously. Performance metrics such as mAP@0.5, Precision, and Recall were used to evaluate model convergence.

After training, the best-performing model weights were exported for real-time inference.

5.10 Real-Time Detection Pipeline:

During runtime, the system continuously captures frames from the input video stream. Each frame undergoes pre-processing before being passed to the trained YOLOv8 model for detection.

The detection output includes:

- Bounding box coordinates
- Class predictions
- Confidence scores

Non-Maximum Suppression (NMS) is applied to remove duplicate detections. Only predictions above a confidence threshold are forwarded to the tracking module.

This detection pipeline operates sequentially for each frame, ensuring consistent monitoring.

CHAPTER-VI

FEEDBACK AND ITERATIVE REFINEMENT

6.1 Feedback Collection from Users

During the initial testing phase of the system, feedback was gathered from faculty members, project mentors, and students who interacted with the exam monitoring application. Users evaluated the system's ability to detect suspicious behavior, the clarity of the video output, and the usefulness of the dashboard interface. Their observations helped identify both strengths and limitations of the model. Many reviewers appreciated the concept of using artificial intelligence to automate exam supervision, while also suggesting improvements in detection accuracy and system responsiveness. In addition, users were asked to comment on the design and usability of the web-based monitoring dashboard developed using Flask. They provided opinions regarding the layout of the interface, the readability of the status indicators, and the usefulness of the color-coded labels that distinguish between normal and suspicious behavior. Many participants appreciated the idea of integrating artificial intelligence with real-time video analysis to automate parts of the examination monitoring process. They noted that such a system could reduce the burden on invigilators and help in identifying potential malpractice more efficiently. However, some reviewers also pointed out certain limitations, such as occasional misclassification of normal behavior as suspicious and the need for improved detection under different lighting conditions or crowded exam settings. A few users suggested enhancing the dataset and further refining the model to increase the accuracy and consistency of the predictions.

6.2 Evaluation of Detection Performance

Based on the feedback received, the detection performance of the system was carefully evaluated. Some users observed that certain suspicious actions were not always detected correctly, while in a few cases normal behavior was occasionally flagged as cheating. This evaluation highlighted the need to improve the dataset quality and adjust the model parameters. Performance metrics such as accuracy, confidence scores, and detection consistency were analyzed to better understand how the model behaved under different conditions. While the overall performance of the detection model was encouraging, some users reported that certain suspicious actions were not always detected with complete accuracy. In a few situations, rapid or subtle movements related to potential cheating behavior were not recognized by the system, which indicated that the model still required additional training data to better understand such patterns. On the other hand, there were also instances where normal student actions, such as looking sideways or adjusting posture, were occasionally classified as suspicious behavior. These cases revealed the possibility of false positives, where the system mistakenly identifies normal activity as cheating. Such observations highlighted the importance of refining the dataset, improving annotation quality, and optimizing the model's parameters to reduce misclassification.

6.3 Model Refinement and Parameter Tuning

To improve the reliability of the system, several refinements were implemented in the model configuration and detection logic. Threshold values, confidence levels, and tracking parameters were adjusted to reduce false detections. The suspicion scoring mechanism was also improved

so that repeated suspicious actions gradually increase the cheating score while normal behavior decreases it over time. These adjustments helped create a more balanced system that minimizes false alarms while maintaining effective cheating detection. One of the key improvements involved adjusting the confidence threshold values used by the object detection model. Confidence thresholds determine the minimum probability required for the system to classify a detected object into a particular category. By fine-tuning these thresholds, the system was able to filter out weak or uncertain detections and focus only on predictions that met a reliable confidence level. Additionally, tracking parameters associated with the object tracking algorithm were optimized to maintain consistent identification of each student across multiple frames. Proper tracking ensured that every detected individual retained a unique identification number throughout the monitoring process, preventing frequent changes in identity and improving the accuracy of behavior analysis over time.

6.4 Interface and System Improvements

User feedback also contributed to improvements in the system interface and monitoring dashboard. The video display, student status indicators, and color-coded bounding boxes were refined to make the monitoring process clearer for invigilators. Additional visual indicators such as normal and cheating status boxes were introduced to provide quick and easy identification of suspicious students. These enhancements improved the usability and readability of the system for real-time monitoring. Another significant improvement involved refining the student status indicators displayed on the monitoring dashboard. The system was designed to track each detected student with a unique identification number, and the interface now clearly displays the status associated with each student in real time. Color-coded bounding boxes were introduced and refined to visually distinguish between different types of behavior detected by the model. For example, green bounding boxes indicate normal student behavior, while red bounding boxes highlight suspicious or cheating activity. The presence of an invigilator is also displayed using a different color to avoid confusion. These color-coded indicators allow invigilators to quickly interpret the monitoring results without needing to read detailed information.

6.5 Continuous Testing and System Validation

After implementing the refinements, the system was tested again in multiple scenarios to validate the improvements. Continuous testing helped ensure that the changes made to the model and interface effectively enhanced the system’s performance. The updated system showed better detection stability and improved tracking of individual students. This iterative process of feedback, modification, and testing played a crucial role in developing a more accurate and reliable exam monitoring solution. The testing process also involved reviewing the evidence recording feature and the live monitoring dashboard to confirm that they functioned correctly after the updates. The system successfully captured short video clips as evidence whenever the cheating threshold was reached, providing valuable proof that could be reviewed later if needed. Meanwhile, the monitoring dashboard displayed real-time updates of student statuses and detection results, enabling invigilators to observe the system’s performance clearly. These tests demonstrated that the improvements made to the interface and detection logic enhanced both usability and functionality.

CHAPTER-VII

TESTING AND VALIDATION

7.1 System Testing Environment

Before deploying the proposed exam monitoring system, it was tested in a controlled environment to evaluate its functionality and performance. The testing setup consisted of a computer system running the Python-based application along with the required libraries such as OpenCV, Flask, and the Ultralytics YOLO framework. A mobile phone camera was used as the video source through an IP webcam stream, which simulated a surveillance camera inside an examination hall. The trained YOLO model was integrated with the system to detect and classify behaviors such as normal activity, cheating behavior, and invigilator presence. The testing environment allowed the developers to verify whether all modules of the system, including video capture, object detection, tracking, and dashboard display, were working together properly. Conducting the testing in a controlled setup helped ensure that the system could process live video input and produce detection results in real time.

7.2 Functional Testing of System Modules

Functional testing was performed to verify that each component of the system worked according to its intended purpose. The system consists of several modules, including video acquisition, object detection, behavior classification, student tracking, suspicion score calculation, and evidence recording. Each module was tested individually to ensure that it performed its specific task correctly. For example, the video acquisition module was tested to confirm that it could successfully capture frames from the IP camera without interruptions. The object detection module was evaluated to verify that the trained model could correctly identify the different classes such as cheating, invigilator, and normal behavior. Similarly, the tracking module was tested to confirm that each detected student was assigned a unique identification number that remained consistent across frames. This functional testing ensured that all parts of the system were operating reliably before integrating them into the final monitoring application.

7.3 Performance Testing of the Detection Model

Performance testing was conducted to measure how accurately and efficiently the detection model performed when analyzing video frames. The trained YOLO model was evaluated using various performance indicators such as detection accuracy, confidence scores, and processing speed. The system was tested with different video inputs containing multiple students performing normal and suspicious activities. The objective of this testing was to determine whether the model could correctly identify cheating-related behavior while avoiding misclassification of normal actions. The system's ability to process frames at an acceptable speed was also examined to ensure that the monitoring process remained close to real time. The results of the performance testing demonstrated that the detection model was capable of identifying behaviors with satisfactory accuracy while maintaining consistent tracking of students across multiple frames.

7.4 Real-Time Monitoring and Tracking Validation

To further validate the effectiveness of the proposed system, real-time monitoring tests were conducted using live camera input. During these tests, the system continuously analyzed incoming video frames and displayed the detection results on the monitoring dashboard. The object tracking algorithm was observed to ensure that it could maintain stable tracking of individual students even when they moved within the camera frame. The suspicion scoring mechanism was also evaluated to confirm that repeated suspicious activities gradually increased the cheating score while normal behavior reduced the score over time. The system successfully maintained tracking IDs for students and displayed color-coded bounding boxes to represent their detected behavior. This validation confirmed that the system could reliably monitor examination activities in real time.

7.5 System Validation and Result Analysis

The final stage of testing involved validating the overall performance of the system by analyzing the results obtained during various test scenarios. The system's ability to detect suspicious behavior, maintain student tracking, and store evidence videos was carefully examined. Test results indicated that the system was capable of identifying suspicious actions with reasonable accuracy while minimizing false detections through the use of suspicion scoring and threshold mechanisms. The evidence recording feature was also verified to ensure that short video clips were successfully saved whenever cheating behavior exceeded the defined threshold. The results confirmed that the system functions as an effective automated monitoring tool that can assist invigilators in maintaining examination integrity. Overall, the testing and validation process demonstrated that the proposed system is reliable, efficient, and suitable for practical implementation in exam monitoring environments.

7.6 Evidence Recording and Alert System Testing

In addition to detection and tracking validation, the evidence recording feature of the system was also thoroughly tested to ensure that it functions correctly when suspicious behavior is detected. The purpose of this testing was to verify whether the system successfully captures and stores video evidence whenever a student's suspicion score crosses the predefined threshold. During the testing process, various scenarios were simulated where repeated cheating behavior was performed in front of the camera. The system continuously analyzed the video frames and increased the suspicion score whenever cheating behavior was detected. Once the suspicion score exceeded the defined threshold value, the system automatically triggered the evidence recording mechanism. The recorded video clips contained the last few seconds of frames before and during the suspicious activity, which were saved in the designated evidence folder with a unique timestamp and student identification number. The stored video files were later reviewed to ensure that the recorded footage clearly captured the detected behavior and could be used as supporting evidence if required. The testing also verified that the system does not repeatedly save evidence for the same student once it has already been recorded, preventing unnecessary duplication of files.

CHAPTER-VIII

RESULTS AND DISCUSSION

8.1 System Output Screenshots

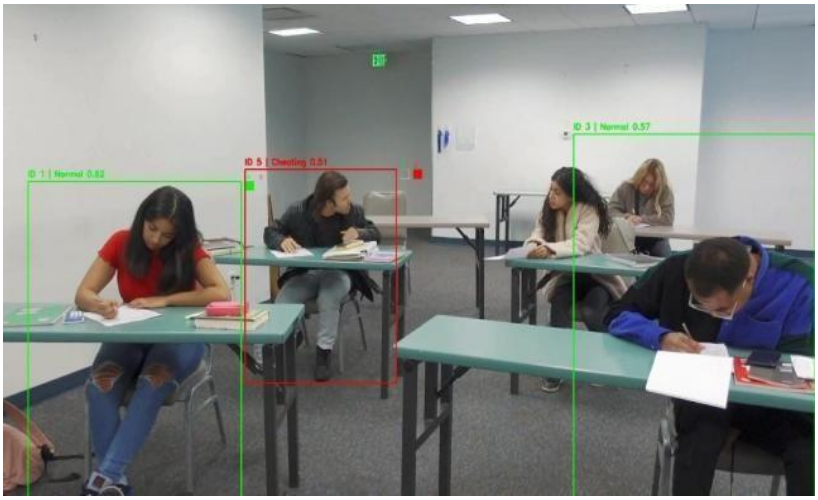

Fig:4 Suspicious and normal

In Figure 4 Images illustrate how the proposed invigilation system identifies and classifies suspicious behaviors during examinations. In one image, a student is observed repeatedly turning towards a neighbouring desk, an action that triggers the system's "Cheating" classification.

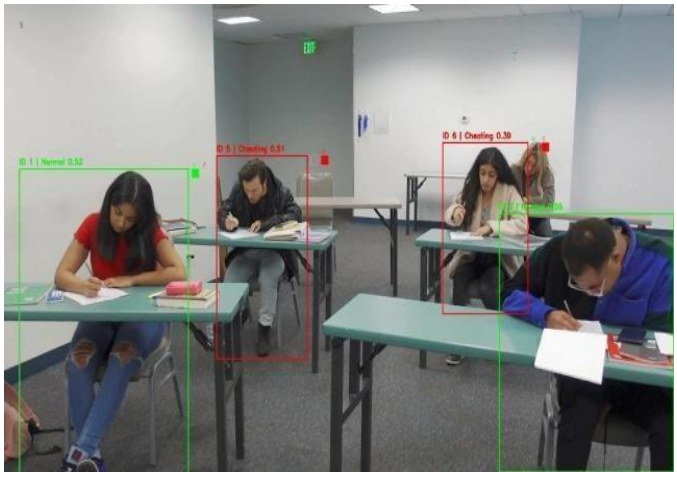

Fig:5 Suspicious and normal

The figure 5 shows two students in close proximity, appearing to exchange materials, which is flagged as abnormal behavior by the system. These visual results confirm that the integration of YOLO for object detection and Byte Track for continuous tracking enables accurate real-time identification of malpractice. The bounding boxes and color-coded labels in the frames not only help in monitoring but also provide clear visual evidence for review and assessment, supporting the objective of maintaining exam integrity with minimal human intervention.

8.2 Evaluation Metrics

In this section, the results that were obtained after training the YOLOv8 model using the custom dataset will be discussed. The model was subjected to testing in order to determine the extent to which the model can identify various activities of the students, such as normal activities, suspicious activities, and the activities of invigilators in the classroom. In order to determine the performance of the model, various measures were investigated, including the precision, recall, F1 score, and the average average precision of the model. Additionally, the confusion matrix and training loss were utilized in determining the performance of the model.

To measure the effectiveness of the trained model, several important evaluation metrics were considered. These metrics included precision**,** recall**,** F1 score**,** and mean Average Precision (mAP)**,** which are commonly used to evaluate the performance of object detection systems. Precision indicates how accurately the model predicts positive detections, meaning how many of the detected cheating or normal behaviors are actually correct. Recall measures the ability of the model to detect all relevant instances of a particular class, indicating how many actual behaviors present in the data were successfully identified by the model. The F1 score provides a balanced measure that combines both precision and recall to give an overall indication of the model's classification performance. In addition, mean Average Precision (mAP) was calculated to evaluate the detection accuracy of the model across all classes and intersection-over-union (IoU) thresholds, providing a comprehensive assessment of how well the model performs in object detection tasks.

Apart from these evaluation metrics, other analytical tools were also used to better understand the model's behavior during training and testing. The confusion matrix was analyzed to observe how frequently the model correctly classified each class and to identify cases where misclassification occurred. This matrix provided insights into whether normal activities were sometimes incorrectly predicted as suspicious or whether cheating-related actions were missed by the system. Furthermore**,** training and validation loss curves were studied to observe how the model's learning process progressed over time. A gradual decrease in training loss and validation loss indicated that the model was successfully learning patterns from the dataset and improving its detection capability. By examining these results together, it was possible to evaluate both the strengths and limitations of the trained YOLOv8 model and determine whether it was suitable for real-time exam monitoring applications. Overall, the analysis of these performance metrics and validation results demonstrated that the trained model achieved satisfactory detection accuracy and provided a reliable foundation for implementing an automated exam behavior monitoring system.

The model was trained to recognize and classify different types of activities that occur in an examination environment, including normal student behavior, suspicious or cheating-related behavior, and the presence of an invigilator. After completing the training phase, the model was evaluated using a separate validation and testing dataset to determine how effectively it could detect and classify these activities in various scenarios. During the testing process, the trained model analyzed multiple video frames and images to verify whether it could correctly identify the objects and behaviors associated with each class.

8.2.1 F1-Confidence Curve

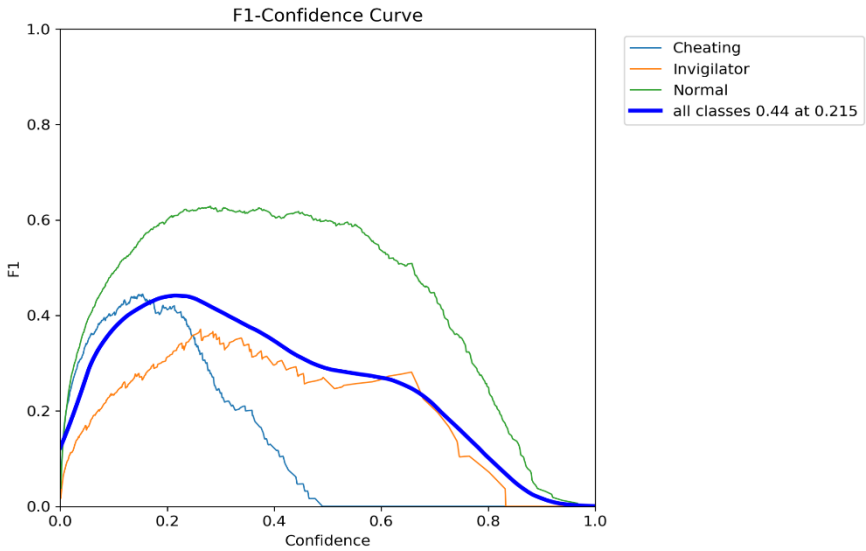

Fig:6 F1-Confidence curve

The F1-Confidence curve shows the effect of the confidence level of the model on the precision and recall of the model. From the results obtained in Figure 4, the Normal category has the highest F1-score compared to the Cheating and Invigilator categories. This shows that the model is more reliable in recognizing students with regular behavior compared to the others.

8.2.2 Precision-Recall Curve

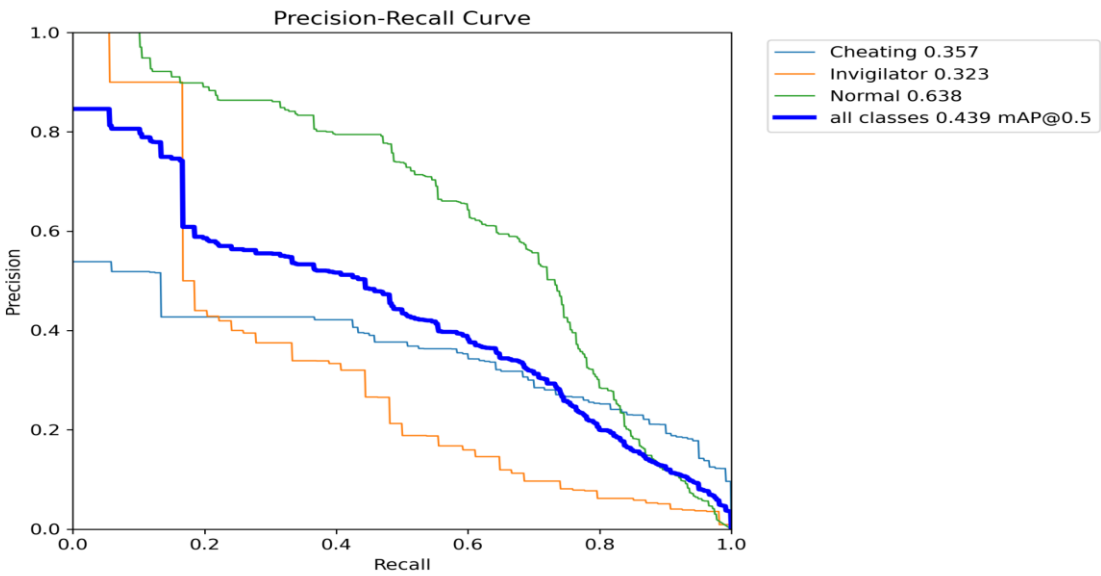

Fig:7 Precision-Recall Curve

The Precision-Recall curve in Figure 5 highlights the balance between precision and recall across classes. The 'Normal' class again shows superior performance with a precision-recall area of 0.638, while the 'Cheating' and 'Invigilator' classes achieve lower scores of 0.357 and 0.323 respectively. The overall mAP@0.5 across all classes is 0.439, which indicates moderate detection capability.

8.2.3 Confusion Matrix

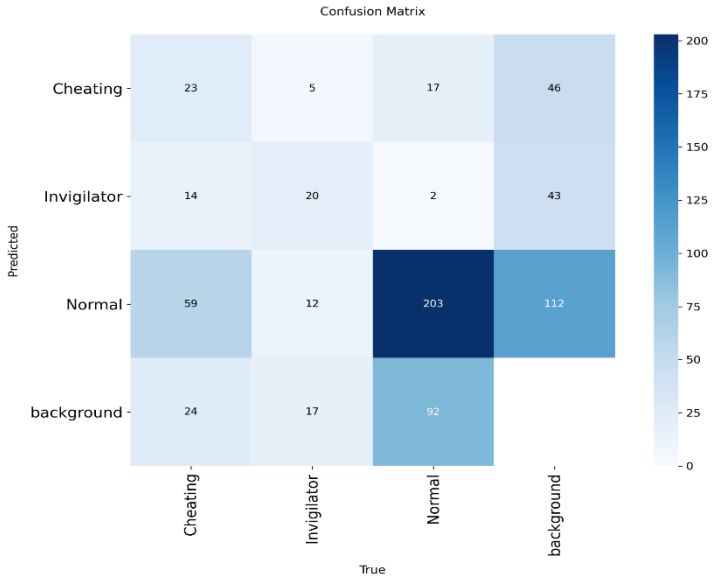

Fig:8 Confusion matrix

The confusion matrix provides a detailed breakdown of correct and incorrect classifications. It shows that the model frequently misclassifies 'Cheating' students as 'Normal', which explains the lower precision and recall values for the 'Cheating' class. Similarly, 'Invigilator' detections also overlap with the 'Normal' category. These findings highlight areas for dataset refinement.

8.3 Comparison with Existing Systems

The proposed AI-based exam monitoring system was also compared with several existing examination monitoring methods to evaluate its effectiveness and practical advantages. Traditionally, examination monitoring in educational institutions relies heavily on manual invigilation, where invigilators supervise students directly in the examination hall to ensure that no unfair practices occur. Although this method is widely used, it has several limitations such as human fatigue, difficulty in monitoring a large number of students simultaneously, and the possibility of missing subtle suspicious activities. In contrast, the proposed system uses artificial intelligence and computer vision techniques to automatically analyze student behavior in real time.

By utilizing a trained YOLOv8 object detection model, the system can continuously monitor the examination environment and detect behaviors related to normal activities, suspicious actions, and the presence of invigilators without constant human observation. This significantly reduces the workload of invigilators and improves the efficiency of monitoring.

Compared with other existing automated monitoring systems, many traditional systems focus only on simple motion detection or require expensive surveillance equipment and complex hardware installations. Some earlier research systems rely on rule-based approaches or basic image processing techniques, which often struggle to accurately detect complex human behaviors in dynamic environments. These systems may also suffer from high false detection rates due to their inability to adapt to different scenarios. In contrast, the proposed system uses a deep learning-based object detection model that is trained on a custom dataset specifically designed for exam monitoring scenarios. This allows the model to better recognize patterns associated with suspicious behavior and distinguish them from normal student movements. Furthermore, the integration of an object tracking algorithm ensures that each student is assigned a unique identification number, allowing the system to monitor individual behavior over time.

Another significant advantage of the proposed system over existing approaches is the implementation of a suspicion scoring mechanism. Instead of immediately labelling a student as cheating based on a single detection, the system gradually increases a suspicion score when repeated suspicious activities are detected. This approach helps reduce false positives and provides a more balanced evaluation of student behavior. Additionally, the system includes an automatic evidence recording feature, which stores short video clips whenever suspicious behavior exceeds the defined threshold. This feature is particularly useful for providing verifiable proof of detected incidents, which is rarely available in traditional monitoring methods.

Moreover, the system provides a web-based monitoring dashboard that displays real-time video streams along with student status indicators. The dashboard uses color-coded bounding boxes and status labels to clearly indicate whether a student's behavior is normal or suspicious. This user-friendly interface allows invigilators to quickly identify potential issues without manually observing every student in the exam hall. Compared to many existing systems that lack interactive visualization tools, this feature significantly improves usability and real-time monitoring efficiency.

Overall, the comparison with existing systems demonstrates that the proposed AI-based exam monitoring system offers several advantages, including automated behavior detection, real-time monitoring, improved accuracy through deep learning, and the ability to record evidence of suspicious activities. While traditional invigilation methods remain important, the integration of intelligent monitoring technologies provides a powerful tool that can enhance examination security and support invigilators in maintaining academic integrity.

8.4 Challenges Faced

The development of the proposed AI-based exam monitoring system involved several technical and practical challenges. The major challenges encountered during the implementation of the project are listed below:

Dataset Collection and Annotation

One of the main challenges was collecting a sufficient dataset that represents different student behaviors in an examination environment. The dataset needed to include images showing normal behavior, suspicious or cheating-related actions, and invigilator presence. In addition, manually labelling the dataset with accurate bounding boxes for training the YOLOv8 model required significant time and effort.

Detection Accuracy and False Predictions

Achieving high detection accuracy was another challenge during the project. Some normal actions such as turning the head or adjusting posture were occasionally misclassified as suspicious behavior. These false detections required adjustments in the model parameters and threshold values to improve the reliability of the system.

Video Quality and Environmental Conditions

The system's performance depended heavily on the quality of the video input. Factors such as poor lighting, low camera resolution, and improper camera placement sometimes affected the detection results. When multiple students appeared in the frame, partial occlusion also made it difficult for the model to clearly detect each individual.

Real-Time Processing Performance

Since the system processes video frames continuously, maintaining real-time performance was a challenge. The system needed to perform detection, tracking, scoring, and video streaming simultaneously, which required efficient processing and optimization to avoid delays.

Integration of Multiple Technologies

The project involved integrating several technologies such as YOLOv8 for object detection, OpenCV for video processing, Byte Track for object tracking, and Flask for the web-based dashboard. Ensuring that all these components worked together smoothly required careful coding, debugging, and system testing.

8.5 Solution and Improvements

The challenges encountered during the development of the AI-based exam monitoring system were addressed through several solutions and improvements. These modifications helped enhance the accuracy, efficiency, and reliability of the system. The key solutions and improvements implemented in the project are listed below:

Improvement of Dataset Quality

To improve detection accuracy, additional images representing different student behaviors were collected and included in the dataset. The dataset was carefully annotated to ensure that bounding boxes were correctly placed around the detected objects. Increasing the dataset diversity helped the YOLOv8 model learn better patterns and improved its ability to distinguish between normal and suspicious activities.

Adjustment of Model Parameters

The confidence threshold, Intersection over Union (IoU) values, and other detection parameters were fine-tuned during the training and testing phases. These adjustments helped reduce false detections and improved the model's ability to correctly classify student behavior in different situations.

Implementation of Suspicion Scoring Mechanism

A suspicion scoring system was introduced to reduce the chances of incorrect cheating detection. Instead of immediately labelling a student as cheating, the system gradually increases a suspicion score when repeated suspicious actions are detected. Normal behavior decreases the score over time, creating a more balanced and reliable decision process.

Optimization for Real-Time Processing

To ensure smooth real-time performance, video frames were resized and processing operations were optimized. Efficient frame handling and resource management helped maintain continuous monitoring without significant delays in detection and tracking.

Enhancement of User Interface and Dashboard

The monitoring dashboard was improved to make it more user-friendly for invigilators. Features such as color-coded bounding boxes, student status indicators, and real-time updates were added to help users easily understand the monitoring results.

CHAPTER-IX

CONCLUSION & FUTURE SCOPE

9.1 CONCLUSION

The increasing need to maintain academic integrity in examination environments has highlighted the limitations of traditional manual invigilation methods. Large examination halls, dynamic student behavior, and limited human supervision make it difficult to ensure fair and reliable monitoring. To address these challenges, this project proposed an AI-powered real-time exam invigilation support system integrating YOLOv8 for object detection and Byte Track for multi-object tracking. The developed system successfully combines fast and accurate object detection with persistent identity tracking to enable continuous monitoring of students during examinations. YOLOv8 effectively detects students, invigilators, and suspicious objects within each video frame, while Byte Track maintains stable Track IDs across consecutive frames, preventing identity switching and tracking loss. This integration ensures reliable long-term behavioral observation in dynamic and crowded examination environments.

One of the key strengths of the proposed framework is the incorporation of temporal suspicious behavior analysis. Instead of relying on single-frame detection, the system evaluates suspicious activity across multiple consecutive frames using a threshold-based mechanism. This significantly reduces false positives and improves the reliability of alerts. The color-coded visualization system further enhances usability by clearly distinguishing normal students (green bounding boxes) from suspicious students (red bounding boxes), enabling quick interpretation by invigilators. Experimental evaluation demonstrates that the system achieves moderate detection accuracy, with stronger performance for the Normal class and comparatively lower accuracy for Cheating and Invigilator classes due to dataset imbalance and visual similarity challenges. Despite these limitations, the integration of Byte Track enhances tracking stability and ensures persistent monitoring across frames.

9.2 FUTURE SCOPE

Although the proposed AI-powered real-time exam invigilation support system demonstrates effective detection, tracking, and suspicious behavior analysis, there are several opportunities for further enhancement and improvement. Future developments can focus on increasing detection accuracy, improving robustness under complex examination conditions, and expanding system capabilities. One of the primary areas for future improvement is dataset expansion and balancing. Increasing the number of annotated images for the Cheating and Invigilator classes can help reduce class imbalance and improve detection accuracy. A more diverse dataset including variations in lighting conditions, seating arrangements, camera angles, and classroom layouts would enhance the model's generalization capability. Another potential enhancement involves integrating pose estimation techniques to better analyze student body posture and gestures. By incorporating body key point detection, the system could more accurately identify suspicious movements such as repeated head turning or unusual hand gestures. This would provide deeper behavioral understanding beyond bounding box detection. Future work may also include gaze direction analysis to monitor where students are looking during examinations. Gaze tracking can help detect frequent side glances or abnormal viewing patterns that may indicate potential malpractice. Integrating gaze analysis with the existing detection and tracking framework could significantly improve behavioral accuracy. Improving robustness under occlusion and crowded conditions is another important direction. Advanced tracking strategies or model fine-tuning can enhance identity stability when students partially block each other. Multi-camera integration could also be explored to provide better coverage in large examination halls and minimize blind spots. The system can be further optimized for low-light and variable illumination environments by applying advanced pre-processing techniques or training with augmented lighting conditions. This would improve performance in diverse classroom settings.

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PUBLICATION

AI-Powered Real-Time Exam Invigilation Support System Using YOLO and Byte Track

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***Abstract*— The problem of cheating during examinations is a major threat to the integrity of examination processes. The purpose of our project is to provide a real-time system of Exam Invigilation using computer vision techniques to improve the efficiency of the examination process. The object detection algorithm used is YOLO (You Only Look Once), which enables the system to detect the students, invigilators, as well as other suspicious objects such as the use of a mobile phone or the use of notes within the examination hall. In addition, the system uses the Byte Track algorithm to perform the object tracking process. The outcome of the project is to provide recorded video or real-time video as input to the system, which is then used to detect cases of cheating during the examination process. The use of the IP WebApp enables the system to capture the activities of students in real-time using the camera of a mobile phone, with the images then being viewed on the PC screen with red boxes indicating the suspicious students and green boxes indicating the normal students. The proposed solution processes the data in real-time with minimal human involvement, which improves the level of scalability, accuracy, and fairness of the exam invigilation process.**

Keywords*— *AI-Powered Invigilation, YOLO, Byte Track, Object Detection, Multi-Object Tracking, Suspicious Behavior Monitoring, Quality Education (SDG-4), Industry Innovation Infrastructure (SDG-9), Peace justice and Strong Institution (SDG-16).

I. INTRODUCTION

Examinations are vital to assess the knowledge and progress of a student. Nevertheless, the increasing rate of malpractices during examinations has turned out to be a major threat to academic integrity. Human invigilators can only monitor a few students at a time. In a large examination hall or an online exam with hundreds of students, it is almost impossible to monitor all malpractices (such as copying from other students, using mobile phones, and whispering answers to students). That is why large-scale malpractices during examinations go undetected using traditional methods. There is an increasing need to develop a more reliable system to ensure fair examination practices using technology. Our project attempts to address this problem by developing an automated surveillance system to monitor examination halls. It is designed to intelligently monitor examination halls and detect malpractices during examinations.

students, detect suspicious activities, and reduce the chances of cheating.

In the proposed system, we are using a powerful object detection technique named YOLO, which is efficient and accurate for object detection. The proposed system can use the YOLO technique to detect students, invigilators, as well as forbidden objects like mobile phones and cheat sheets. In order to improve the efficiency of the proposed system for monitoring activities and detecting cheating attempts, we plan to use a powerful technique named Byte Track, which is efficient for tracking multiple objects. The technique can be used to track students and objects to detect suspicious activities.

The rest of the paper is presented as follows: In Section II, a brief overview of the existing literature regarding the existing approaches to the usage of AI in the examination monitoring system is presented .In Section III, the methodology is presented. In this section, a brief description of data acquisition, input data pre-processing, and the real-time monitoring system using the YOLO algorithm and the Byte Track algorithm is presented. In Section IV, a brief description of the evaluation of the proposed methodology is presented. In Section V, a brief description of the conclusion of the paper is presented. In this section, a brief description of the major contributions, future scope, and scope of improvement in the automated examination invigilation system is presented.

II. RELATED WORK

Arumugam [1] proposed a cheating detection model based on deep learning techniques and ensemble techniques such as YOLO, faster R-CNN, and multimodal fusion techniques. It was reported to have a high accuracy compared to other traditional models. However, it was challenged by the computational overhead. Rajasekar and Vijayarani [2] proposed a cheating detection system based on a deep learning technique and computer vision techniques. It was reported to have promising results. However, it was challenged by the issue of crowded rooms. Nagajothi et al. [3] proposed a prototype system based on artificial intelligence to detect cheating in exams based on the use of the YOLO algorithm and facial recognition techniques. It was

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